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United States Patent	12413054
Kind Code	B2
Date of Patent	September 09, 2025
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Stripping tool

Abstract

A stripping tool includes a fixed handle (10), a fixed clamp plate (20), a movable handle (30), a movable clamp plate (40), a cutting/pulling mechanism (50) and a driving operation mechanism (60). The driving operation mechanism (60) includes a pushing assembly (61), a pressing assembly (62) and an action member (63). One end of the pushing assembly (61) is connected to the moveable handle (30) and another end is moveably connected to the action member (63). The pressing assembly (62) is fixed to the fixed handle (10). One end of the action member (63) is connected to the cutting/pulling mechanism (50) and another end is elastically pressed by the pressing assembly (62). When the movable handle (30) rotates toward the fixed handle (10), the pushing assembly (61) moves the action member (63) and the cutting/pulling mechanism (50) toward the pressing assembly (62) to implement wire stripping.

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Appl. No.:	18/104296
Filed:	February 01, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240258774 A1	Aug. 01, 2024

Publication Classification

Int. Cl.: H02G1/12 (20060101)

U.S. Cl.:

CPC **H02G1/1212** (20130101);

Field of Classification Search

CPC: B25B (5/02); B25B (5/04); B25B (5/06); B25B (7/00); B25B (7/02); B25B (7/04); B25B (7/12); B25B (7/22); H02G (1/12); H02G (1/1212); H02G (1/1236)

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Background/Summary

BACKGROUND

Technical Field

(1) The disclosure relates to a technology of a hand tool, particularly to a stripping tool.

Related Art

(2) General plumber and electrician works such as maintenance and piping usually require to strip outer insulated layers of various cables or wires to expose the cores in the cables or wires, so as to be ready for subsequent electric connection. Thus, there are wire strippers in the market for removing outer insulated layers of cables or wires without damaging inner cores to smooth the stripping process.

(3) A wire stripper includes a first handle, a second handle, a first pressing member, a second pressing member and a stripping assembly. The first handle, the second handle, the first pressing member and the second pressing member are pivoted together. The stripping assembly is installed between the first pressing member and the second pressing member, and acts with the first handle and the second handle. Therefore, the first pressing member, the second pressing member and the stripping assembly may be driven to perform wire stripping by pressing the first handle and the

second handle.

(4) However, such a wire stripper has the following problems. Related elements are disposed in a limited space, so its moment arm cannot be effectively increased and causes considerable laboriousness in the using process. Also, because too many components are to be assembled, the manufacturing of the wire strippers is complicated, the structure is unstable and the cost is high.

(5) In view of this, the inventors have devoted themselves to the above-mentioned related art, researched intensively and cooperated with the application of science to try to solve the above-mentioned problems. Finally, the invention which is reasonable and effective to overcome the above drawbacks is provided.

SUMMARY

(6) An object of the disclosure is to provide a stripping tool, which may achieve the function of labor-saving operation and stable operation.

(7) To accomplish the above object, the disclosure provides a stripping tool, which includes a fixed handle, a fixed clamp plate, a movable handle, a movable clamp plate, a cutting/pulling mechanism, and a driving operation mechanism. The fixed clamp plate is connected to the fixed handle. The movable handle is connected pivotally to the fixed handle and formed under the fixed clamp plate. The movable clamp plate is connected pivotally to the fixed handle and correspondingly formed above the fixed clamp plate. The cutting/pulling mechanism is disposed between the fixed clamp plate and the movable clamp plate. The driving operation mechanism includes a pushing assembly, a pressing assembly, and an action member. One end of the pushing assembly is connected to the moveable handle and another end of the pushing assembly is moveably connected to the action member. The pressing assembly is fixed to the fixed handle. One end of the action member is connected to the cutting/pulling mechanism and another end of the action member is elastically pressed by the pressing assembly. When the movable handle is triggered to rotate toward the fixed handle, the pushing assembly moves the action member and the cutting/pulling mechanism toward the pressing assembly to implement wire stripping.

(8) The disclosure further has the following function. By the strip-shaped sliding trough, the cutting/pulling mechanism may not shift when each cutting knife is used to cut a wire or cable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an assembled perspective view of the stripping tool of the disclosure;

(2) FIG. 2 is an exploded view of the stripping tool of the disclosure;

(3) FIG. 3 is an exploded view of the stripping tool of the disclosure from another viewpoint;

(4) FIG. 4 is an exploded view of the driving operation mechanism tool of the disclosure;

(5) FIG. 5 is an assembled cross-sectional view of the stripping tool of the disclosure;

(6) FIG. 6 is a cross-sectional view of the stripping tool of the disclosure which is performing cutting; and

(7) FIG. 7 is a cross-sectional view of the stripping tool of the disclosure which is performing pulling.

DETAILED DESCRIPTION

(8) The technical contents of this disclosure will become apparent with the detailed description of embodiments accompanied with the illustration of related drawings as follows. It is intended that the embodiments and drawings disclosed herein are to be considered illustrative rather than restrictive.

(9) Please refer to FIGS. 1-5. The disclosure provides a stripping tool, which is a wire stripper in the embodiment and includes a fixed handle **10**, a fixed clamp plate **20**, a movable handle **30**, a movable clamp plate **40**, a cutting/pulling mechanism **50** and a driving operation mechanism **60**.

(10) The fixed clamp plate **20** is extended from the fixed handle **10**. The movable handle **30** is connected pivotally to the fixed handle **10** and formed under the fixed clamp plate **20**. The movable clamp plate **40** is connected pivotally to the fixed handle **10** and correspondingly formed above the fixed clamp plate **20**. The cutting/pulling mechanism **50** is disposed between the fixed clamp plate **20** and the movable clamp plate **40**. The driving operation mechanism **60** includes a pushing assembly **61**, a pressing assembly **62** and an action member **63**. One end of the pushing assembly **61** is connected to the moveable handle **30** and another end of the pushing assembly **61** is moveably connected to the action member **63**. The pressing assembly **62** is fixed to the fixed handle **10**. One end of the action member **63** is connected to the cutting/pulling mechanism **50** and another end of the action member **63** is elastically pressed by the pressing assembly **62**.

(11) In an embodiment, the fixed handle **10** includes a head portion **11**, a stem portion **12** downward and aslant extended from the head portion **11**, and a throat portion **13** downward and aslant extended from the head portion **11** and deflecting from the stem portion **12**. The movable handle **30** is penetrated through by a first shaft **A1** to implement the pivotal connection between the movable handle **30** and the throat portion **13** of the fixed handle **10**.

(12) In an embodiment, the movable clamp plate **40** and the fixed handle **10** are penetrated through by a second shaft **A2** to implement the pivotal connection therebetween. The right side of the movable clamp plate **40** is extended with a driven arm **41**. A side of the head portion **11** of the fixed handle **10** is disposed with a blocking plate **111** used for limiting the rotational stroke of the driven arm **41**. A front end of the fixed clamp plate **20** is installed with lower clamping teeth **21**, and a front end of the movable clamp plate **40** is connected with upper clamping teeth **42**. The upper clamping teeth **42** are arranged corresponding to the lower clamping teeth **21** to jointly clamp a wire or cable. Also, the inside of the lower clamping teeth **21** is disposed with a lower trough **22** and a scale **23**.

(13) In an embodiment, the movable handle **30** and one end of the pushing assembly **61** are penetrated through by a third shaft **A3** so as to make the movable handle **30** be able to drive the action member **63** and the cutting/pulling mechanism **50** to move together.

(14) In an embodiment, the cutting/pulling mechanism **50** is disposed between the fixed clamp plate **20** and the movable clamp plate **40**, and includes a lower pressing member **51**, an upper pressing member **52**, a stopping member **53**, an elastic body **54** and a bolt **55**. The upper pressing member **52**, the lower pressing member **51** and the action member **63** are penetrated through by the bolt **55**. The stopping member **53** is movably disposed on the lower pressing member **51** and arranged corresponding to the scale **23**. The elastic body **54** is elastically clamped between the lower pressing member **51** and the upper pressing member **52**. The front end of the lower pressing member **51** is disposed with a lower cutting knife **511**, and the front end of the upper pressing member **52** is disposed with an upper cutting knife **521** arranged corresponding to the lower cutting knife **511**.

(15) In an embodiment, a guiding trough **112** is formed in the head portion **11** of the fixed handle **10** (as shown in FIG. 5) to allow the action member **63** to move in or move out. The action member **63** is disposed with a strip-shaped sliding trough **631** and an oriented trough **632**. The strip-shaped sliding trough **631** is arranged longitudinally and the oriented trough **632** is arranged transversely. A first fixing plug **B1** penetrates through the oriented trough **632** and is fixed to the fixed handle **10** so as to limit the action member **63** to reciprocate transversely. Please refer to FIG. 4. In an embodiment, the pushing assembly **61** includes two pushing rods **611** and an elastic body **615**. The strip-shaped sliding trough **631** of the action member **63** and each pushing rod **611** are penetrated through by a spindle **612** so as to implement the movable connection of each pushing rod **611** and the action member **63**. A part of the spindle **612** outside each pushing rod **611** is connected with a roller **613** to perform the functions of stopping, vibration absorption and noise reduction by the contact between the roller **613** and the first fixing plug **B1** when the movable handle **30** returns. Also, an inserting rod **614** penetrates through each pushing rod **611** to provide positioning of an end

of the elastic body **615**. Furthermore, the inside of the movable handle **30** has a pressing plate **32** as shown in FIG. **5**. The elastic body **615** includes a ring **616** and a first arms **617** and a second arm **618** extended from the ring **616**. The third shaft **A3** penetrates through the movable handle **30**, each pushing rod **611** and the ring **616** of the elastic body **615**. Each first arm **617** elastically abuts against the inserting rod **614**, and the second arm **618** elastically abuts against the pressing plate **32** of the movable handle **30**, thereby allowing the movable handle **30** to return to the original position. The first arm **617** may be designed as a single piece.

(16) In an embodiment, the pressing assembly **62** is installed correspondingly at a joint between the head portion **11** and the stem portion **12** of the fixed handle **10**, and is positioned by a second fixing plug **B2** penetrating therethrough. The pressing assembly **62** includes a base **621** and an elastic element **625**. The base **621** is formed with a receiving trough **622**. The base **621** is disposed with a bolt **623**. The rear of the receiving trough **622** is formed with a block **624**. The elastic element **625** includes a ring **626** and a first elastic arms **627** and a second elastic arm **628** extended from the ring **626**. The ring **626** is received in the receiving trough **622**, and is positioned by the insert **623** penetrating through the base **621** and the ring **626**. An end of each first elastic arm **627** abuts against the blocking block **624**. An end of the second elastic arm **628** abuts against the action member **63** to allow the action member **63** to return. The first elastic arm **627** may be designed as a single piece.

(17) In an embodiment, the stripping tool of the disclosure further includes a dividing knife **70**. The top of the movable handle **30** is provided with an embedding trough **31**. The dividing knife **70** includes a fixed knife **71** installed on the throat portion **13** of the fixed handle **10** and a movable knife **72** installed on the embedding trough **31** of the movable handle **30** corresponding to the fixed knife **71**. The movable knife **72** may cut a wire or cable with respect to the fixed knife **71** with the inserting of the first shaft **A1** when rotating the movable handle **30**.

(18) In an embodiment, the stripping tool of the disclosure further includes a cutting depth adjustment mechanism **80** disposed between the movable clamp plate **40** and the cutting/pulling mechanism **50**. The opening width between the lower pressing member **51** and the upper pressing member **52** may be limited by the adjustment and shift of the cutting depth adjustment mechanism **80**, so as to be able to match different specifications of various wires or cables to be adaptively adjusted.

(19) Please refer to FIGS. **6** and **7**. When operating, a cable or wire **9** is placed between the upper clamping teeth **42** and lower clamping teeth **21**, and an end face of the cable or wire **9** abuts against the stopping member **53**. When the movable handle **30** is triggered to rotate toward the fixed handle **10**, the movable clamp plate **40** may move the upper pressing member **52** toward the lower pressing member **51** to close. By clamping the cable or wire **9** with the upper clamping teeth **42** and the lower clamping teeth **21**, and cutting the covering layer of the cable or wire **9** with the upper cutting knife **521** and the lower cutting knife **511**, the spindle **612** of each pushing rod **611** shifts toward the top of the strip-shaped sliding trough **631** and the first fixing plug **B1** is located at the right end of the oriented trough **632** of the action member **63**, so as to prevent the cutting/pulling mechanism **50** from shifting during cutting the cable or wire **9** as shown in FIG. **6**.

(20) When continuously rotating the movable handle **30** toward the fixed handle **10**, if its action force is greater than the elastic force of the elastic element **625** of the pressing assembly **62**, each pushing rod **611** may drive the action member **63** and the cutting/pulling mechanism **50** to move toward the pressing assembly **62**, and the upper cutting knife **521** and the lower cutting knife **511** may pull out the covering layer of the cable or wire **9** as shown in FIG. **7**. The above structure may delay the time of the covering layer of the cable or wire **9** being pulled out, after the cutting/pulling mechanism **50** cut that, to make the whole action process more stable and smoother.

(21) While this disclosure has been described by means of specific embodiments, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope and spirit of this disclosure set forth in the claims.

Claims

1. A stripping tool comprising: a fixed handle (10); a fixed clamp plate (20), connected to the fixed handle (10); a movable handle (30), connected pivotally to the fixed handle (10), and disposed under the fixed clamp plate (20); a movable clamp plate (40), connected pivotally to the fixed handle (10), and correspondingly disposed above the fixed clamp plate (20); a cutting/pulling mechanism (50), disposed between the fixed clamp plate (20) and the movable clamp plate (40); a driving operation mechanism (60), comprising a pushing assembly (61), a pressing assembly (62) and an action member (63), wherein one end of the pushing assembly (61) is connected to the moveable handle (30) and another end of the pushing assembly (61) is moveably connected to the action member (63), the pressing assembly (62) is fixed to the fixed handle (10), and one end of the action member (63) is connected to the cutting/pulling mechanism (50) and another end of the action member (63) is elastically pressed by the pressing assembly (62); and a first fixing plug (B2), wherein the pressing assembly (62) comprises a base (621) and an elastic element (625), the base (621) is fixed to the fixed handle (10) by the first fixing plug (B2), the base (621) comprises a bolt (623) and a receiving trough (622), the receiving trough (622) is defined by a blocking block (624), the elastic element (625) comprises a ring (626) and a first elastic arm (627) and a second elastic arm (628) extended from the ring (626), the ring (626) is received in the receiving trough (622), the elastic element (625) is positioned by the bolt (623) passing through the base (621) and the ring (626), the first elastic arm (627) abuts against the blocking block (624), and the second elastic arm (628) abuts against the action member (63); wherein when the movable handle (30) is triggered to rotate toward the fixed handle (10), the pushing assembly (61) moves the action member (63) and the cutting/pulling mechanism (50) toward the pressing assembly (62) to implement wire stripping.
2. The stripping tool of claim 1, further comprising a first shaft (A1), wherein the fixed handle (10) comprises a head portion (11), a stem portion (12) downward extended from the head portion (11) and a throat portion (13) extended from the head portion (11) and deflecting from the stem portion (12), and the movable handle (30) is connected pivotally with the throat portion (13) through the first shaft (A1).
3. The stripping tool of claim 2, further comprising a dividing knife (70), wherein the movable handle (30) comprises an embedding trough (31), the dividing knife (70) comprises a fixed knife (71) installed on the throat portion (13) and a movable knife (72) installed on the embedding trough (31) corresponding to the fixed knife (71), and the first shaft (A1) passes through the fixed knife (71) and the movable knife (72).
4. The stripping tool of claim 1, further comprising a second shaft (A2), wherein the movable clamp plate (40) is connected pivotally to the fixed handle (10) through the second shaft (A2), the movable clamp plate (40) comprises a driven arm (41) extended therefrom, and the fixed handle (10) comprises a blocking plate (111) to limit a rotational stroke of the driven arm (41).
5. The stripping tool of claim 1, further comprising a third shaft (A3), wherein the movable handle (30) is configured to drive the action member (63) and the cutting/pulling mechanism (50) to move through one end of the pushing assembly (61) being passed through by the third shaft (A3).
6. The stripping tool of claim 1, further comprising a second fixing plug (B1), wherein the action member (63) comprises an oriented trough (632), and the second fixing plug (B1) passes through the oriented trough (632) and is fixed to the fixed handle (10) to limit a moving direction of the action member (63).
7. The stripping tool of claim 1, wherein the pushing assembly (61) comprises two pushing rods (611) and a spindle (612), the action member (63) comprises a strip-shaped sliding trough (631), and each pushing rod (611) is movably connected to the action member (63) by the spindle (612) passing through the strip-shaped sliding trough (631) and each pushing rod (611).

8. A stripping tool comprising: a fixed handle (10); a fixed clamp plate (20), connected to the fixed handle (10); a movable handle (30), connected pivotally to the fixed handle (10), and disposed under the fixed clamp plate (20); a movable clamp plate (40), connected pivotally to the fixed handle (10), and correspondingly disposed above the fixed clamp plate (20); a cutting/pulling mechanism (50), disposed between the fixed clamp plate (20) and the movable clamp plate (40); a driving operation mechanism (60), comprising a pushing assembly (61), a pressing assembly (62) and an action member (63), wherein one end of the pushing assembly (61) is connected to the moveable handle (30) and another end of the pushing assembly (61) is moveably connected to the action member (63), the pressing assembly (62) is fixed to the fixed handle (10), and one end of the action member (63) is connected to the cutting/pulling mechanism (50) and another end of the action member (63) is elastically pressed by the pressing assembly (62); and wherein when the movable handle (30) is triggered to rotate toward the fixed handle (10), the pushing assembly (61) moves the action member (63) and the cutting/pulling mechanism (50) toward the pressing assembly (62) to implement wire stripping; wherein the pushing assembly (61) comprises two pushing rods (611) and a spindle (612), the action member (63) comprises a strip-shaped sliding trough (631), and each pushing rod (611) is movably connected to the action member (63) by the spindle (612) passing through the strip-shaped sliding trough (631) and each pushing rod (611); wherein the pushing assembly (61) further comprises two rollers (613), and each roller (613) is adapted to sheath the spindle (612) and located outside each pushing rod (611).

9. A stripping tool comprising: a fixed handle (10); a fixed clamp plate (20), connected to the fixed handle (10); a movable handle (30), connected pivotally to the fixed handle (10), and disposed under the fixed clamp plate (20); a movable clamp plate (40), connected pivotally to the fixed handle (10), and correspondingly disposed above the fixed clamp plate (20); a cutting/pulling mechanism (50), disposed between the fixed clamp plate (20) and the movable clamp plate (40); a driving operation mechanism (60), comprising a pushing assembly (61), a pressing assembly (62) and an action member (63), wherein one end of the pushing assembly (61) is connected to the moveable handle (30) and another end of the pushing assembly (61) is moveably connected to the action member (63), the pressing assembly (62) is fixed to the fixed handle (10), and one end of the action member (63) is connected to the cutting/pulling mechanism (50) and another end of the action member (63) is elastically pressed by the pressing assembly (62); and a third shaft (A3), wherein the pushing assembly (61) further comprises an inserting rod (614) and an elastic body (615), the movable handle (30) comprises a pressing plate (32), the inserting rod (614) passes through each pushing rod (611), the elastic body (615) comprises a ring (616) and a first arm (617) and a second arm (618) extended from the ring (616), the third shaft (A3) passes through the movable handle (30), each pushing rod (611) and the ring (616), the first arm (617) elastically abuts against the inserting rod (614), and the second arm (618) elastically abuts against the pressing plate (32); wherein when the movable handle (30) is triggered to rotate toward the fixed handle (10), the pushing assembly (61) moves the action member (63) and the cutting/pulling mechanism (50) toward the pressing assembly (62) to implement wire stripping; wherein the pushing assembly (61) comprises two pushing rods (611) and a spindle (612), the action member (63) comprises a strip-shaped sliding trough (631), and each pushing rod (611) is movably connected to the action member (63) by the spindle (612) passing through the strip-shaped sliding trough (631) and each pushing rod (611).
